KNN

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Imports

In [97]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split, StratifiedKFold
from sklearn.neighbors import KNeighborsClassifier as KNN
from sklearn.metrics import accuracy_score, ConfusionMatrixDisplay, confusion_ma
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import GridSearchCV
from sklearn.pipeline import Pipeline
import seaborn as sns
from sklearn.metrics import confusion_matrix
from sklearn.metrics import roc_curve, auc, RocCurveDisplay
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
```

Load Data

In [98]:
```
spaceship_data = pd.read_csv('final_data.csv')
```

split to x and y

In [99]:
```
y = spaceship_data["Transported"].astype(int)
x = spaceship_data.drop(columns=["Transported", "GroupId"])
```

In [100…
```
knn_pipe = Pipeline([
    ("knn", KNN())
])
```

Train-Test Split

In [101…
```
x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=42, stratify=y
)
```

In [102…
```
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
```

KNN Model

In [103…

```
param_grid = {
    "knn__n_neighbors": list(range(1, 40, 2)),
    "knn__weights": ["uniform", "distance"],
    "knn__metric": ["euclidean", "manhattan"]
}

grid = GridSearchCV(
    knn_pipe,
    param_grid,
    cv=cv,
    scoring="accuracy",
    n_jobs=-1
)

grid.fit(x_train, y_train)
best_model = grid.best_estimator_
```

results

In [104…

```
best_model = grid.best_estimator_
```

In [105…

```
y_pred_train  = best_model.predict(x_train)
```

In [106…

```
train_acc = accuracy_score(y_train, y_pred_train)
```

אחוז דיוק על test

In [107…

```
print("="*40)
print("BEST KNN — TRAIN SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {train_acc*100:.2f}%")
print("="*40)
```

```
========================================
BEST KNN — TRAIN SET PERFORMANCE
========================================
Accuracy : 79.60%
========================================
```

טבלה שמציגה את הטבלה להצגת 5 המודלים הכי טובים שנלמדו

In [108…

```
results = pd.DataFrame(grid.cv_results_)

summary = results[
    [
        "param_knn__n_neighbors",
        "param_knn__weights",
        "param_knn__metric",
        "mean_test_score",
        "std_test_score",
        "rank_test_score"
    ]
].sort_values("rank_test_score")
```

```
summary.head(5)
```

Out[108…

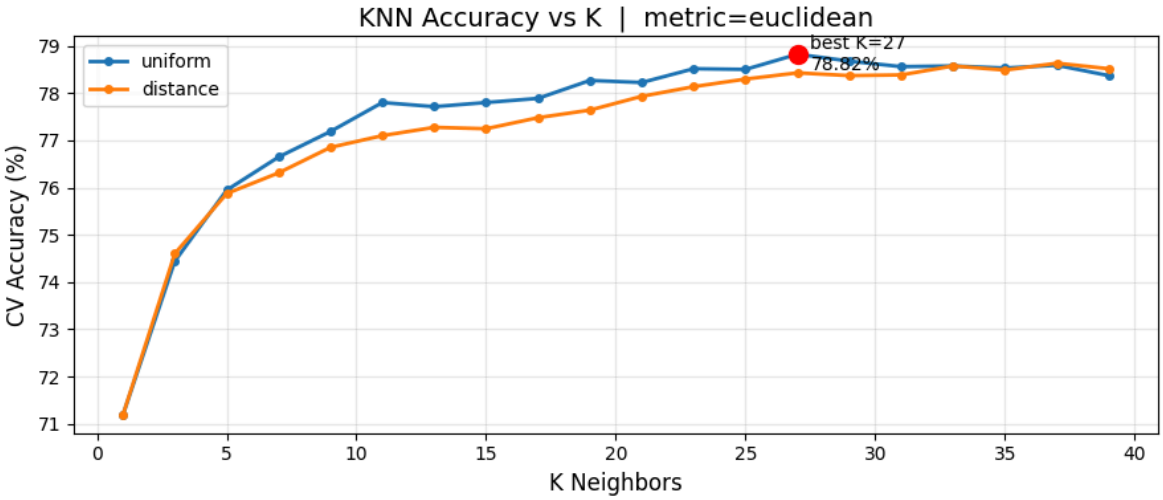
	param_knn__n_neighbors	param_knn__weights	param_knn__metric	mean_test_score
26	27	uniform	euclidean	0.788231
76	37	uniform	manhattan	0.787500
74	35	uniform	manhattan	0.787500
68	29	uniform	manhattan	0.786916
28	29	uniform	euclidean	0.786770

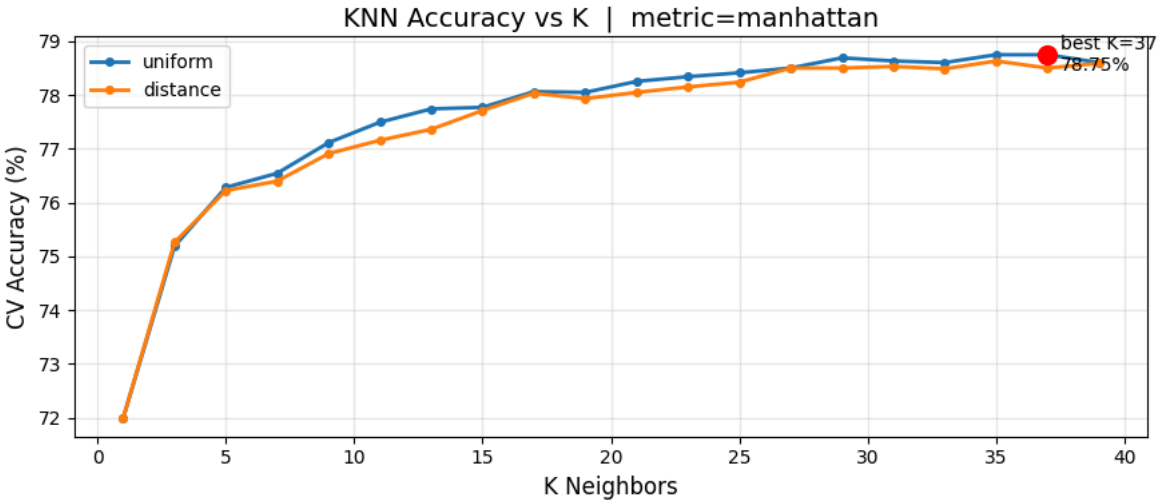
גרף שמראה את אחוז הדיוק של המודל הטוב ביותר לשאר המודלים

In [109…

```
results = pd.DataFrame(grid.cv_results_)

summary = results[[
    "param_knn__n_neighbors",
    "param_knn__weights",
    "param_knn__metric",
    "mean_test_score",
    "std_test_score"
]].copy()

summary["K"] = summary["param_knn__n_neighbors"].astype(int)
summary["weights"] = summary["param_knn__weights"].astype(str)
summary["metric"] = summary["param_knn__metric"].astype(str)

for metric in sorted(summary["metric"].unique()):
    sub = summary[summary["metric"] == metric].sort_values("K")

    plt.figure(figsize=(9, 4))

    for w in ["uniform", "distance"]:
        curve = sub[sub["weights"] == w].sort_values("K")
        plt.plot(
            curve["K"],
            curve["mean_test_score"] * 100,
            marker="o",
            linestyle="-",
            linewidth=2,
            markersize=4,
            label=w
        )

    # best (הכי גבוה מתוך שתי העקומות) נקודה אדומה
    best_row = sub.loc[sub["mean_test_score"].idxmax()]
    best_k = int(best_row["K"])
    best_acc = float(best_row["mean_test_score"] * 100)

    plt.scatter(best_k, best_acc, color="red", s=100, zorder=5)
    plt.text(best_k + 0.5, best_acc, f"best K={best_k}\n{best_acc:.2f}%", va="ce

    plt.xlabel("K Neighbors", fontsize=12)
    plt.ylabel("CV Accuracy (%)", fontsize=12)
    plt.title(f"KNN Accuracy vs K  |  metric={metric}", fontsize=14)
```

```
    plt.grid(True, alpha=0.3)
    plt.legend()
    plt.tight_layout()
    plt.show()
```





חישוב Precision, Recall, F1 מתוך ה-CM

In [110…

```
cm_train = confusion_matrix(y_train, y_pred_train)

TN, FP, FN, TP = cm_train.ravel()

precision = TP / (TP + FP)
recall    = TP / (TP + FN)
f1        = 2 * (precision * recall) / (precision + recall)

print(f"Precision: {precision:.3f}")
print(f"Recall:    {recall:.3f}")
print(f"F1 Score:  {f1:.3f}")
```

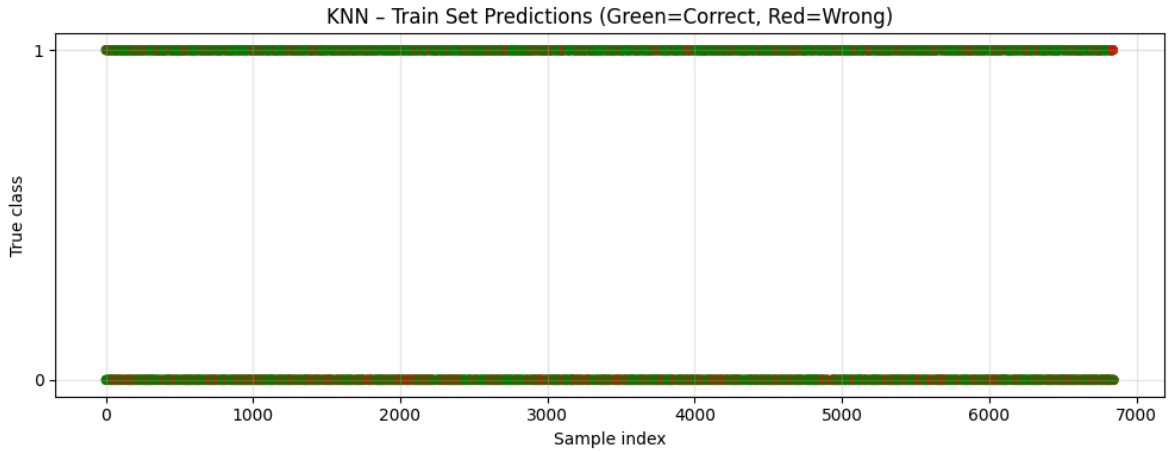
```
Precision: 0.771
Recall:    0.845
F1 Score:  0.807
```

מטריצת ערבול למודל המוצלח ביותר TRAIN

In [111…

```
correct = (y_pred_train == y_train)

# צבעים: ירוק = נכון, אדום = שגוי
```

```
colors = np.where(correct, "green", "red")

plt.figure(figsize=(10,4))
plt.scatter(range(len(y_train)), y_train, c=colors, s=25, alpha=0.7)

plt.yticks([0,1])
plt.xlabel("Sample index")
plt.ylabel("True class")
plt.title("KNN – Train Set Predictions (Green=Correct, Red=Wrong)")
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()

print(f"KNN Train Accuracy: {100*train_acc:.2f}%")
print(f"Errors: {np.sum(~correct)} / {len(correct)}")
```

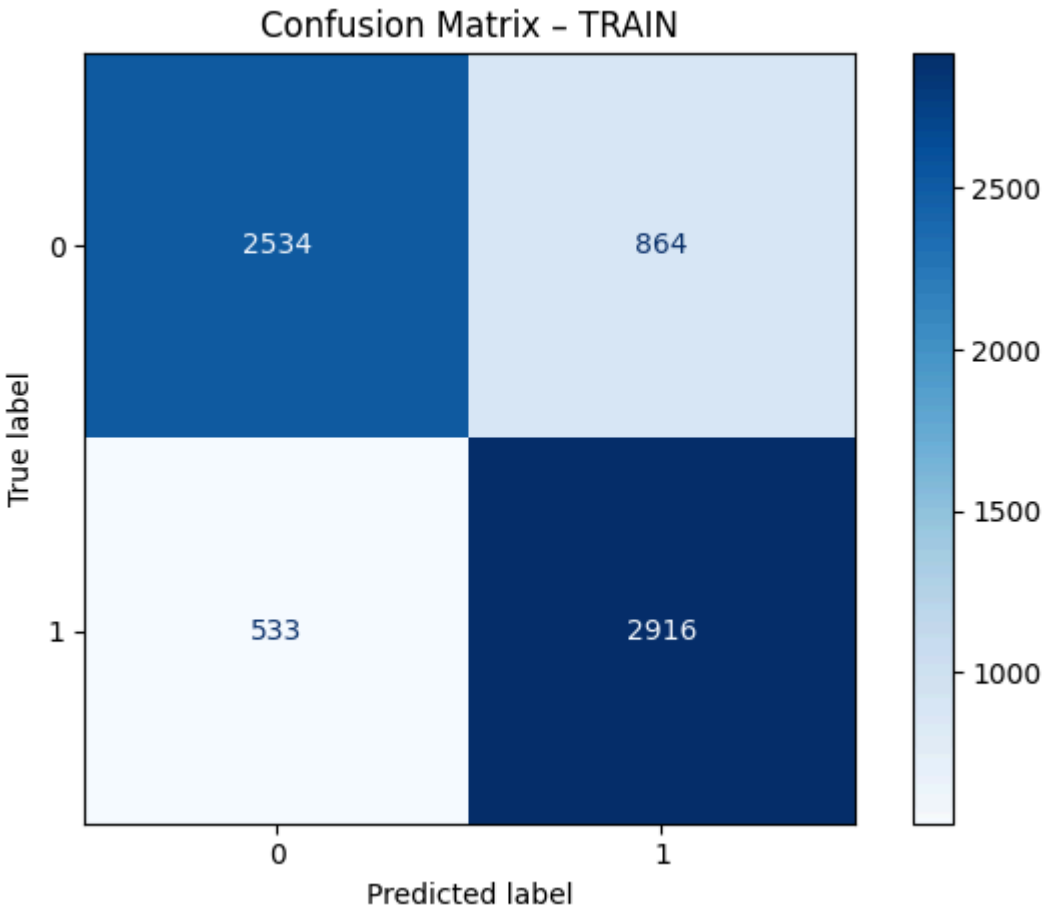



```
KNN Train Accuracy: 79.60%
Errors: 1397 / 6847
```

In [112…

```
disp_train = ConfusionMatrixDisplay(
    confusion_matrix=cm_train
)

plt.figure(figsize=(6,6))
disp_train.plot(cmap="Blues", values_format="d")
plt.title("Confusion Matrix – TRAIN")
plt.tight_layout()
plt.show()
```

```
<Figure size 600x600 with 0 Axes>
```


Confusion Matrix – TRAIN
2534
864
533
2916
0
1
True label
0
1
Predicted label
2500
2000
1500
1000